

## On group operations other than $xy$ or $yx$

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**1. Introduction.** Professor E. MARCZEWSKI suggested recently a notion of *weak isomorphism* of algebraic systems (see „Definition” below). In this paper we solve a problem raised by him, concerning the weak isomorphism of groups.

In the course of the solution we show the existence of groups  $G$  which have the following

*Property* (\*): There is a binary operation  $x \circ y$  in  $G$ , other than  $xy$  or  $yx$ , i. e. a word

$$x \circ y = x^{k_1} y^{l_1} x^{k_2} y^{l_2} \dots x^{k_r} y^{l_r} \quad (k_i, l_i \text{ integers})$$

not identically equal in  $G$  to  $xy$  or to  $yx$ , such that

- (i) the elements of  $G$  form a group  $G_\circ$  under the operation  $x \circ y$ ,
- (ii) the operation  $xy$  is a word in  $G_\circ$ , i. e. there are integers  $m_1, \dots, m_s$  and  $n_1, \dots, n_s$  such that

$$xy = [x]_\circ^{m_1} \circ [y]_\circ^{n_1} \circ \dots \circ [x]_\circ^{m_s} \circ [y]_\circ^{n_s}$$

holds identically for all  $x, y$  in  $G$  ( $[x]_\circ^m$  denotes the  $m$ -th power of  $x$  with respect to the multiplication  $\circ$  in  $G_\circ$ ).

G. HIGMAN and B. H. NEUMANN [1] have proposed the following problem: "Is there any binary operation in a group  $G$ , other than  $xy^{-1}$  or  $yx^{-1}$  and their transposes, in terms of which all group operations can be expressed?" Our examples of groups with Property (\*) answer this question. For, it is clear that if  $G$  has Property (\*), then  $x \circ y^{(-1)}$  where by  $y^{(-1)}$  we mean the inverse element of  $y$  with respect to the  $\circ$  multiplication is an operation required by G. HIGMAN and B. H. NEUMANN.

As has been recently shown by HANNA NEUMANN [3], if  $G$  is a free group, then any binary operation in  $G$  which is associative is of one of the following types:  $a, x, y, xay, yax$  where  $a$  is any constant element of  $G$ . This covers an unpublished result of K. URBANIK that a free group does not possess Property (\*). It follows from our Theorem 1 (see the Remark) that a free nilpotent group of class 2 does not possess Property (\*) either.

We prove below that if  $G$  is a periodic nilpotent group of class 2 and  $G$  has Property (\*), then the groups  $G$  and  $G_\circ$  are isomorphic (Theorem 2). We do not know whether our assumptions are essential and we would like to suggest the following

PROBLEM. If  $G$  has Property  $(*)$ , are  $G$  and  $G_\circ$  isomorphic?

We now proceed to define the notion of weak isomorphism of two algebraic systems. By an *algebraic system*, or shortly *algebra*, we mean a pair  $(A, F)$  composed of a set  $A$  and a family  $F$  of operations (functions) of finitely many variables defined on  $A$  and taking values in  $A$ . For every positive integer  $n$  we define the class  $\mathbf{A}^{(n)}$  of *algebraic operations of  $n$  variables* as the smallest class of operations such that

(j) the identity operations  $e_j^n(x_1, \dots, x_n)$  defined by

$$e_j^n(x_1, \dots, x_n) = x_j \text{ for all } x_1, \dots, x_n,$$

belong to  $\mathbf{A}^{(n)}$ ,

(jj) if  $f \in F$  is an operation of  $m$  variables and  $g_i(x_1, \dots, x_n)$ ,  $i = 1, \dots, m$ , are operations belonging to  $\mathbf{A}^{(n)}$ , then the operation

$$f(g_1(x_1, \dots, x_n), \dots, g_m(x_1, \dots, x_n))$$

belongs to  $\mathbf{A}^{(n)}$ .

We call  $\mathbf{A} = \bigcup_{n=1}^{\infty} \mathbf{A}^{(n)}$  the class of algebraic operations of the algebra  $(A, F)$  (cf. [2]).

We shall identify a group  $G$  with the algebra  $(A, F)$  where  $A$  is the set of elements of  $G$  and  $F$  is the class composed of the unary operation  $x^{-1}$  and the binary operation  $xy$ , i. e.  $F = \{x^{-1}, xy\}$ . Then it is easily seen that  $\mathbf{A}$  is the class of all operations  $f(x_1, \dots, x_n)$  which are given by words in  $G$ , i. e. which are of the form

$$(1) \quad f(x_1, \dots, x_n) = x_{i_1}^{m_1} x_{i_2}^{m_2} \dots x_{i_l}^{m_l}$$

where  $m_1, \dots, m_l$  are arbitrary integers,  $i_1, \dots, i_l \in \{1, \dots, n\}$ , and  $n = 1, 2, \dots$ .

*Definition.* Let  $\mathcal{A}_1 = (A_1, F_1)$  and  $\mathcal{A}_2 = (A_2, F_2)$  be two algebras and let  $\mathbf{A}_1, \mathbf{A}_2$  be the corresponding classes of algebraic operations. We say, following E. MARCZEWSKI, that a one-to-one mapping  $\tau$  of the set  $A_1 \cup \mathbf{A}_1$  onto the set  $A_2 \cup \mathbf{A}_2$  is a *weak isomorphism* of  $\mathcal{A}_1$  onto  $\mathcal{A}_2$  if

$$(k) \quad A_1 \tau = A_2, \mathbf{A}_1^{(n)} \tau = \mathbf{A}_2^{(n)} \text{ for every } n,$$

$$(kk) \quad \text{if } f \in \mathbf{A}_1^{(n)}, \text{ then } [f(a_1, \dots, a_n)]\tau = (f\tau)(a_1\tau, \dots, a_n\tau) \\ \text{for every } a_1, \dots, a_n \text{ in } A_1.$$

If the algebras  $\mathcal{A}_1$  and  $\mathcal{A}_2$  coincide, then we call  $\tau$  a *weak automorphism* of  $\mathcal{A} [= \mathcal{A}_1 = \mathcal{A}_2]$ .

A weak isomorphism  $\tau$  of the algebras  $\mathcal{A}_1$  and  $\mathcal{A}_2$  is an isomorphism in the usual sense if  $F_1\tau = F_2$ . A weak automorphism of  $\mathcal{A}$  is an automorphism in the usual sense if it is the identity mapping on  $F$ .

EXAMPLE. Let  $G = (A, \{x^{-1}, xy\})$  be a group. We define a weak automorphism  $\tau$  of  $G$  by

$$(k_1) \quad x\tau = x^{-1} \text{ for every } x \text{ in } A,$$

$$(k_2) \quad \text{if } f \in \mathbf{A}^{(n)} \text{ is of the form (1),}$$

$$(f\tau)(x_1, \dots, x_n) = x_{i_l}^{m_l} x_{i_{l-1}}^{m_{l-1}} \dots x_{i_1}^{m_1}.$$

We call this weak automorphism *natural*.

**PROBLEM (E. MARCZEWSKI):** Do there exist two groups  $G$  and  $H$  such that there is a weak isomorphism of  $G$  onto  $H$  which is neither an isomorphism (in the usual sense) nor a combination of an isomorphism with the natural weak automorphism of one of the groups?

We give in the sequel a positive answer to this question. In fact, we prove that there is a finite group  $G$  which admits a weak automorphism  $\tau$  such that  $\tau$  is neither an automorphism nor a combination of an automorphism with the natural weak automorphism.

**2. Results.** In the sequel we deal only with nilpotent groups of class 2. It is well known that if  $G$  is nilpotent of class 2, then every word

$$x^{k_1}y^{l_1}x^{k_2}\dots x^{k_r}y^{l_r} \quad (k_i, l_i \text{ integers})$$

is identically equal to a word of the form  $x^ay^b[x, y]^k$  where the integers  $a, b$  and  $k$  depend only on  $k_1, l_1, \dots, k_r, l_r$ . Thus in particular the operation  $x \circ y$  considered in Property (\*) must be of the form  $x^ay^b[x, y]^k$ . We shall prove the following two theorems.

**Theorem 1.** *If  $G$  is a nilpotent group of class 2, then a binary operation  $x \circ y$  in  $G$  has properties (i) and (ii) (cf. (\*)) if and only if*

$$(2) \quad x \circ y = xy[x, y]^k$$

where  $k$  is any integer such that  $2k+1$  is prime to the exponent  $n$  of the derived subgroup  $G'$  of  $G$ .

Then, if and only if  $m$  is an integer such that  $k+m(2k+1)$  is divisible by  $n^*$ , we have that

$$(3) \quad xy = x \circ y \circ [x, y]_{\circ}^m$$

where  $[x, y]_{\circ}^m$  is the  $m$ -th power of the commutator of the elements  $x, y$  with respect to the multiplication  $\circ$  in  $G_{\circ}$ .

**REMARK.** This theorem answers the problem of HIGMAN and NEUMANN (quoted above) in the affirmative. Another consequence of Theorem 1 is that a free nilpotent group of class 2 does not possess Property (\*). For if  $\Gamma$  is free nilpotent of class 2 and an operation  $x \circ y = xy[x, y]^k$  in  $\Gamma$  has properties (i) and (ii), then clearly this operation will have these properties in any homomorphic image  $G$  of  $\Gamma$ . But in the solution of MARCZEWSKI's problem below we define, for every  $n > 1$ , a group  $G$  with two generators which is nilpotent of class 2 and such that the exponent of its derived subgroup is  $n$ . We conclude, by Theorem 1, that  $2k+1$  is prime to every  $n > 1$ , i. e.  $k=0$  or  $-1$ . Therefore  $x \circ y = xy$  or  $yx$ , and these are the only two operations in  $\Gamma$  which have the properties (i) and (ii).

**Theorem 2.** *If  $G$  is a periodic nilpotent group of class 2 which has Property (\*), then the groups  $G$  and  $G_{\circ}$  are isomorphic.*

Postponing the proofs of these theorems to the next section we give now the solution of MARCZEWSKI's problem.

\*) The existence of such an integer  $m$  follows from the fact that  $2k+1$  is prime to  $n$ .

We observe first that, for every integer  $n > 1$ , there is a finite group  $G$  which is nilpotent of class 2 and such that the exponent of the derived subgroup  $G'$  is  $n$ . In fact, let  $C$  be the cyclic group of order  $n$  and let the group  $G$  be defined as the splitting extension of the direct product  $C \times C = \{(a, b) | a, b \in C\}$  by the automorphism  $\alpha$

$$(a, b)^\alpha = (a, ab) \quad \text{for all } a, b \text{ in } C.$$

Then it is easily checked that  $G' = \{(1, a) | a \in C\}$ . Hence  $G'$  is in the centre of  $G$ , and  $G'$  has exponent  $n$ . (For our previous application note that the two elements  $\alpha, (a, 1)$  generate  $G$  whenever  $a$  is a generator of  $C$ ).

Now let  $k$  be any integer such that  $k \not\equiv 0, -1 \pmod{n}$  and  $(2k+1, n) = 1$  (e. g.  $k=1$  if  $n=4$ ). Then the operation  $x \circ y = xy[x, y]^k$  is different from the operations  $xy$  and  $yx$ , and, by Theorem 1,  $x \circ y$  has the properties (i) and (ii). Hence  $G$  has Property (\*). By Theorem 2, there is an isomorphism  $\varphi$  of  $G$  onto  $G_\circ$ , i. e. a mapping  $\varphi$  such that

$$(4) \quad (xy)\varphi = x\varphi \circ y\varphi \quad \text{for all } x, y \text{ in } G.$$

To define a weak automorphism of  $G$ , we consider  $G$  as the algebra  $(A, \{x^{-1}, xy\})$  where  $A$  is the set of elements of  $G$ . Denoting by  $\mathbf{A}^{(n)}$  the class of algebraic operations of  $n$  variables and by  $\mathbf{A}$  the class of algebraic operations, we define the mapping  $\tau$  of  $A \cup \mathbf{A}$  onto itself as follows

$$(k'_1) \quad x\tau = x\varphi \quad \text{for all } x \text{ in } A,$$

$$(k'_2) \quad \text{if } f \in \mathbf{A}^{(n)} \text{ is of the form (1), then}$$

$$(f\tau)(x_1, \dots, x_n) = [x_{i_1}]_\circ^{m_1} \circ [x_{i_2}]_\circ^{m_2} \circ \dots \circ [x_{i_t}]_\circ^{m_t}$$

where, as in (ii),  $[x]_\circ^m$  denotes the  $m$ -th power of  $x$  in  $G_\circ$ . Then  $\tau$  is a weak automorphism of  $G$ . The mapping  $\tau$  is well defined, for if  $f, g \in \mathbf{A}^{(n)}$  are such that  $f=g$  holds identically in  $G$ , then  $f\tau=g\tau$  holds identically in  $G_\circ$ , because  $G$  and  $G_\circ$  are isomorphic. Hence  $f=g$  implies  $f\tau=g\tau$ . By the same argument,  $f\tau=g\tau$  implies  $f=g$ , and thus  $\tau$  is one-to-one on  $\mathbf{A}$ . We also have that  $\tau$  is one-to-one on  $A$ , because  $\tau$  is an isomorphism of  $G$  onto  $G_\circ$ . From (3) it follows that  $\tau$  maps  $\mathbf{A}$  onto  $\mathbf{A}$  and thus, by (4),  $\tau$  is a weak automorphism. But  $\tau$  is not an automorphism nor a combination of an automorphism with the natural weak automorphism, because the  $\tau$ -image of the operation  $xy$ , i. e. the operation  $x \circ y = xy[x, y]^k$ , is different from both  $xy$  and  $yx$ .

**3. Proofs of the Theorems.** Before proving the theorems we wish to recall some well known identities valid in any nilpotent group of class 2. If  $G$  is a nilpotent group of class 2, then for any  $x, y, z$  in  $G$  and any integer  $n$  the following identities hold:

$$(5) \quad (xy)^n = x^n y^n [y, x]^{\frac{n(n-1)}{2}}$$

$$(6) \quad [xy, z] = [x, z][y, z], \quad [x, yz] = [x, y][x, z],$$

hence

$$(7) \quad [x^n, y] = [x, y^n] = [x, y]^n.$$

PROOF OF THEOREM 1. Let  $G$  be a nilpotent group of class 2 and let  $x \circ y$  be an operation in  $G$  which has the properties (i) and (ii) (cf. (\*)). Then, as we observed at the beginning of the previous section

$$x \circ y = x^a y^b [x, y]^k$$

for all  $x, y$  in  $G$  and some integers  $a, b, k$ . Let 1 be the unit element of  $G$ . By the above equality,  $1 \circ 1 = 1$ , hence 1 is also the unit element of  $G_\circ$ . Therefore

$$x = x \circ 1 = x^a, \quad y = 1 \circ y = y^b$$

hold identically in  $G$ , and we have (2). It follows immediately from (2) that, for every integer  $n$ , the  $n$ -th power of any element  $x$  with respect to the operation  $\circ$  is just  $x^n$ . Also, the commutator  $[x, y]_\circ$  in  $G_\circ$  can be easily expressed in terms of the commutator  $[x, y]$ . In fact, a simple application of (2), (6), and (7) shows that

$$(8) \quad [x, y]_\circ = [x, y]^{2k+1}.$$

Hence  $[[x, y]_\circ, z]_\circ = [[x, y]^{2k+1}, z]^{2k+1} = 1$ , i. e.  $G_\circ$  is nilpotent of class 2. From this and from (ii) we infer that for some integers  $a, b, m$

$$xy = x^a \circ y^b \circ [x, y]_\circ^m.$$

The argument used to prove (2) will also serve now to prove that (3) holds identically for all  $x, y$  in  $G$ .

We note that, by (8),  $[x, y]_\circ^m = [x, y]^{m(2k+1)}$ . Substituting this in (3) and using (2) we get

$$xy = xy[x, y]^{k+m(2k+1)}$$

valid for all  $x, y$  in  $G$ . Hence the order of each commutator  $[x, y]$  must divide  $k + m(2k + 1)$  and so, as  $G$  is nilpotent of class 2, the exponent  $n$  of the derived subgroup  $G'$  of  $G$  must also divide  $k + m(2k + 1)$ . It follows that  $2k + 1$  is prime to  $n$ . This completes the proof of the necessity of the conditions.

To prove the sufficiency, we suppose that  $G$  is a nilpotent group of class 2 and that  $k$  is an integer such that  $2k + 1$  is prime to the exponent  $n$  of the derived subgroup  $G'$ . In this case there is an integer  $m$  such that  $k + m(2k + 1)$  is divisible by  $n$ .

To prove (i), we observe first that the operation (2) is associative. In fact, by (7),

$$(x \circ y) \circ z = xyz[x, y]^k [x, z]^k [y, z]^k = x \circ (y \circ z).$$

We also have  $1 \circ x = x \circ 1 = x$  and  $x^{-1} \circ x = x \circ x^{-1} = 1$ , so that the elements of  $G$  form a group  $G_\circ$  under the multiplication  $x \circ y$ .

To prove (ii) we note that, as before, we have (8); whence, applying (2), we deduce that

$$x \circ y \circ [x, y]_\circ^m = xy[x, y]^{k+m(2k+1)}.$$

Using the fact that the exponent of  $G'$  divides  $k + m(2k + 1)$ , we obtain (3). This completes the proof of the theorem.

PROOF OF THEOREM 2. The following well known lemma will be applied in the proof.

**Lemma.** *If  $G$  is a periodic nilpotent group, then  $G$  is the direct product of its Sylow  $p$ -groups.*

We now assume that  $G$  is a periodic nilpotent group of class 2 which has Property (\*). Then, by Theorem 1, the operation  $x \circ y$  is of the form (2) and there is an integer  $m$  such that (3) holds. Moreover, the exponent of the derived subgroup  $G'$  of  $G$  divides the number  $t = k + m(2k + 1)$ .

Let  $t = p_1^{z_1} \dots p_r^{z_r}$  be the factorization of  $t$  into primes and let

$$P = P_1 \times \dots \times P_r$$

be the direct product of the Sylow  $p_i$ -subgroups ( $i = 1, \dots, r$ ) of the group  $G$ . Then, by the lemma,

$$G = P \times Q.$$

Of course each element of  $Q$  has order prime to  $t$ . Moreover, the group  $Q$  is Abelian. For if  $x, y$  are in  $Q$ , then, since the exponent of  $G'$  divides  $t$ ,  $[x, y]$  must belong to  $P$ , so  $[x, y] \in P \cap Q = \{1\}$ . We note that also

$$G_\circ = P_\circ \times Q_\circ,$$

where  $P_\circ$  and  $Q_\circ$  are the groups formed from the elements of  $P$  and  $Q$  respectively under the multiplication  $x \circ y$ .

Let  $s = 2m + 1$ . We have that  $s$  is (prime to  $t$ , for  $2m + 1, m + (2m + 1)k = d$  implies  $d|2m + 1$ , whence  $d|m$ , and so  $d|1$ ). We define now a mapping  $\varphi$  of  $G$  onto  $G_\circ$  by

$$a\varphi = a^s \quad \text{if } a \text{ is in } P,$$

$$b\varphi = b \quad \text{if } b \text{ is in } Q,$$

$$(ab)\varphi = a\varphi \circ b\varphi \quad \text{for any } a \text{ in } P \text{ and } b \text{ in } Q.$$

It is clear that, by  $(s, t) = 1$ ,  $s$  is prime to the orders of the elements of  $P$ , hence  $\varphi$  is one-to-one and onto. We prove that  $\varphi$  is an isomorphism of  $G$  onto  $G_\circ$ . To this purpose it is sufficient to prove that  $\varphi$  is a homomorphism on  $P$ . If  $x, y$  are in  $P$ , we have, by (5),

$$(xy)\varphi = (xy)^s = x^s y^s [y, x]^{\frac{s(s-1)}{2}} = x^s y^s [x, y]^{\frac{s(1-s)}{2}}.$$

On the other hand, by (2) and (7),

$$x\varphi \circ y\varphi = x^s \circ y^s = x^s y^s [x^s, y^s]^k = x^s y^s [x, y]^{s^2 k}.$$

Thus all that remains to prove is that

$$\frac{s(1-s)}{2} \equiv s^2 k \pmod{n}$$

where  $n$  is the exponent of  $G'$ . But this is immediate, for we have  $n|t$  and

$$s^2 k - \frac{s(1-s)}{2} = s \left( sk - \frac{1-s}{2} \right) = s((2m+1)k + m) = st.$$

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